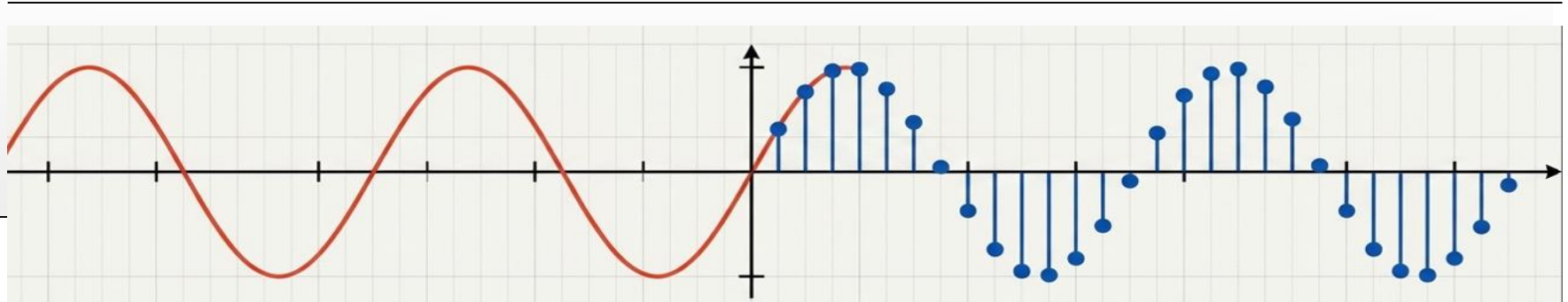


Signal and System

Week 9

Prof. Mehdi Pirahandeh



Continuous vs. Discrete-Time Domains

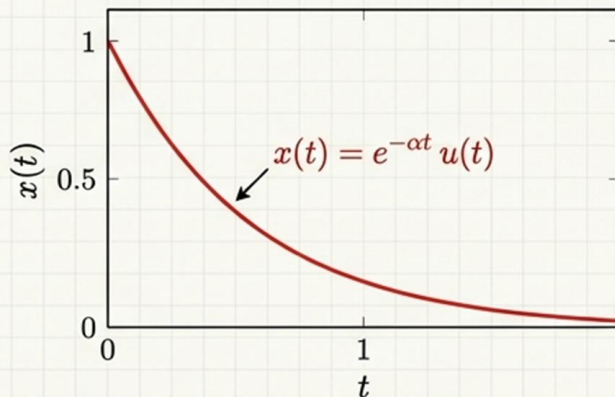
Continuous-Time Domain

Mathematical Representation: Differential equations

Time Intervals: Defined at every instant t

Fourier Transform Type: Continuous-Time Fourier Transform (CTFT)

Visual Characteristic: Smooth curve plot



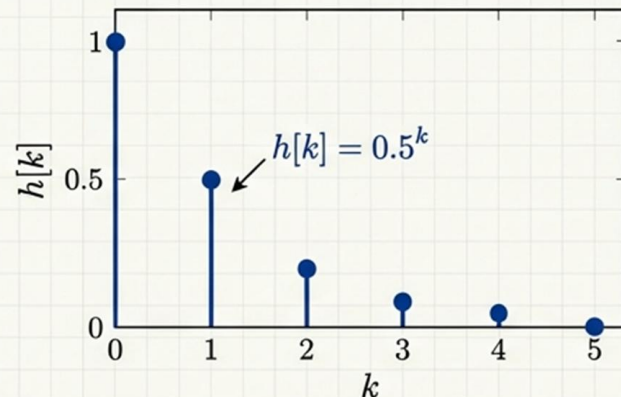
Discrete-Time Domain

Mathematical Representation: Difference equations

Time Intervals: Defined only at discrete intervals n

Fourier Transform Type: Discrete-Time Fourier Series/Transform (DTFS/DTFT)

Visual Characteristic: Stem/lollipop graph



Signal Foundations Dashboard

Periodicity



$$x[n] = x[n+N] \text{ for all } n$$

Defines how the fundamental period N dictates the signal's core frequency components.

Symmetry



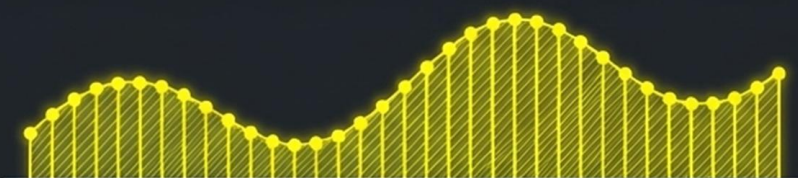
$$\text{Even: } x[n] = x[-n]$$



$$\text{Odd: } x[n] = -x[-n]$$

Simplifies Fourier transforms by isolating cosine (even) and sine (odd) components.

Signal Energy



$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Insight: Total strength of a finite-duration discrete signal.

Signal Power

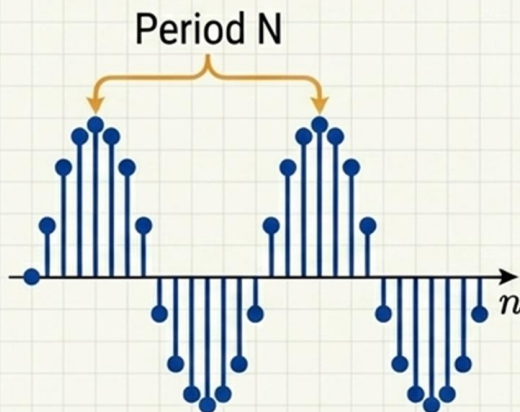


$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

Insight: Average strength of an infinite-duration signal over time.

Fundamental Properties of Discrete-Time Signals

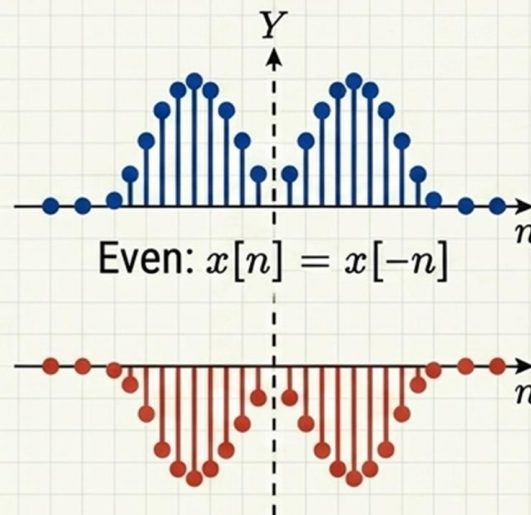
Periodicity



$$x[n] = x[n+N]$$

Period dictates frequency components.

Symmetry

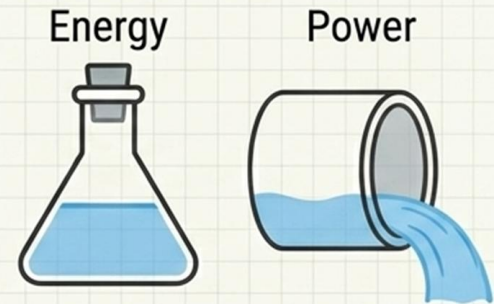


$$\text{Even: } x[n] = x[-n]$$

$$\text{Odd: } x[n] = -x[-n]$$

Simplifies Fourier transforms.

Energy vs. Power



Energy

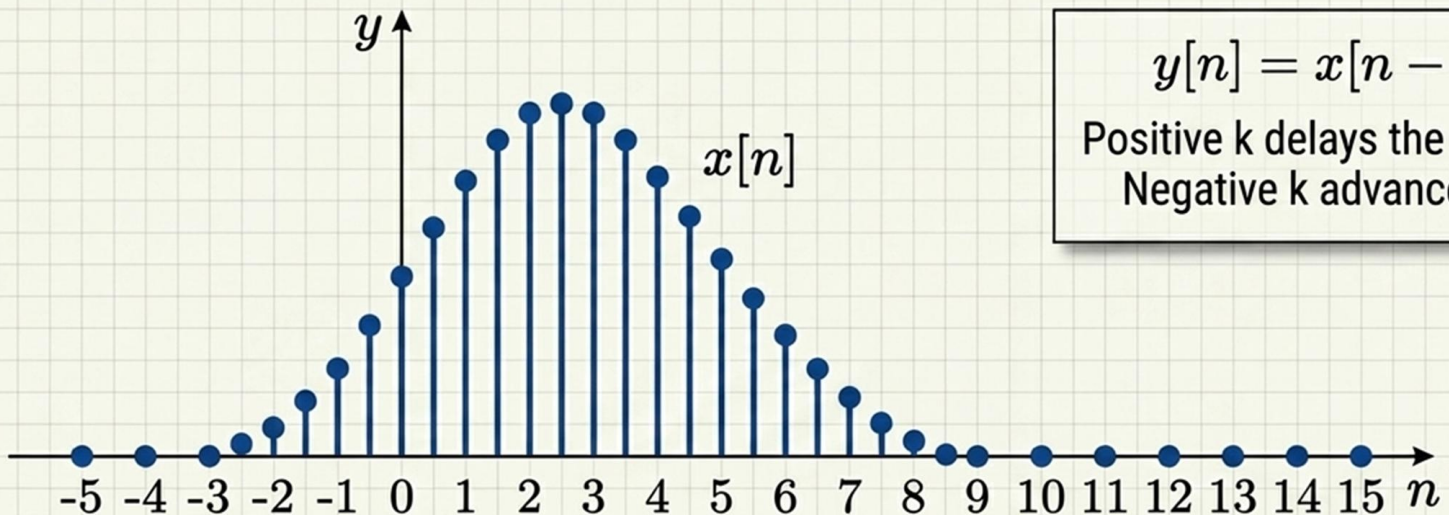
$$E = \sum |x[n]|^2$$

Power

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum |x[n]|^2$$

Anatomy of a Time-Shift

Crucial for echo generation, filter design, and synchronization.



$$y[n] = x[n - k]$$

Positive k delays the signal.
Negative k advances it.

Advance: $k < 0$, shift left

k



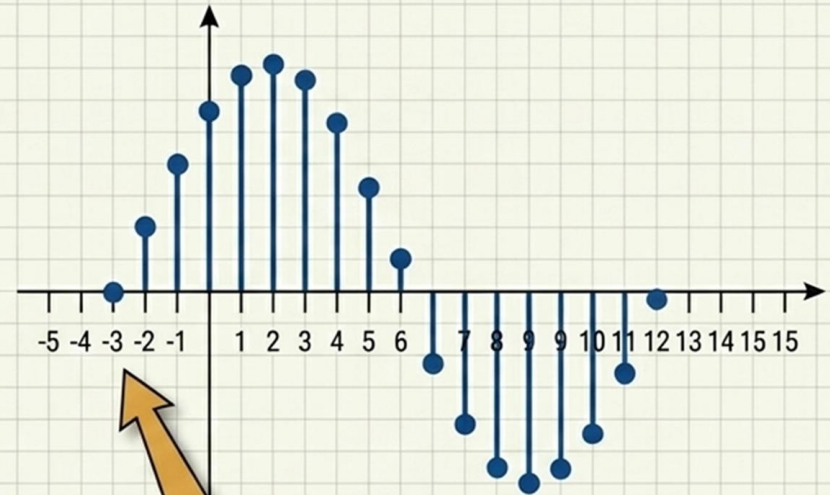
Delay: $k > 0$, shift right

Theory-to-Implementation: Time-Shifting in MATLAB

Syntax Snippet

```
% Define the time shift  
tt = -3; % Negative implies advancing  
xx1 = 0:10; yy1 = sin(xx1);  
  
% Plot shifted discrete data  
stem(xx1 - tt, yy1, 'filled');
```

Visual Output



The shift factor ' $-tt$ ' modifies the independent variable x-axis, not the signal amplitude y.

Crossing the Domain: The Fourier Lens

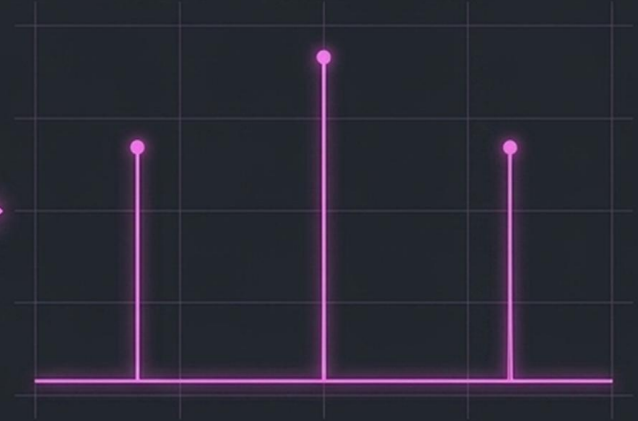
Time Domain



Amplitude over Time. Complex to isolate individual harmonic components.

```
X = fftshift(fft(x, N));
```

Frequency Domain



Magnitude and Phase over Frequency. The complex signal is elegantly unpacked into its fundamental sinusoidal building blocks.

The Frequency Impact: Visualizing Transformations

The Phase Twist - Shifting

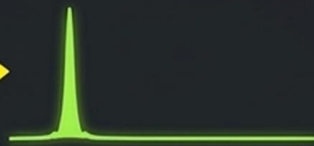


$$x(t - t_0) \rightarrow X(j\omega)e^{-j\omega t_0}$$

Time-shifting does not change the magnitude spectrum; it applies a linear phase shift—a geometric 'twist'—across the frequencies.

The Inverse Scale - Scaling

Part 1: Wide Time Domain



Part 2: Narrow Time Domain

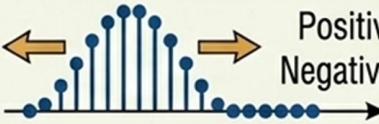
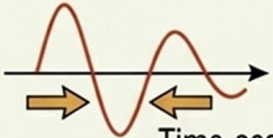
Part 2: Frequency spectrum



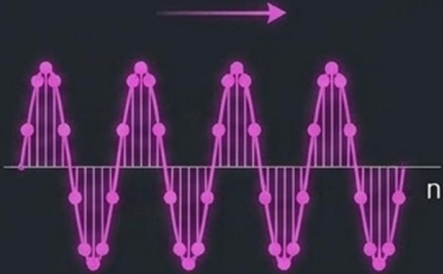
$$x(at) \rightarrow \frac{1}{|a|} X\left(j\frac{\omega}{a}\right)$$

The Inverse Relationship: A signal that is fast/compressed in time is inherently wide/expanded in frequency.

Core Signal Operations Playbook

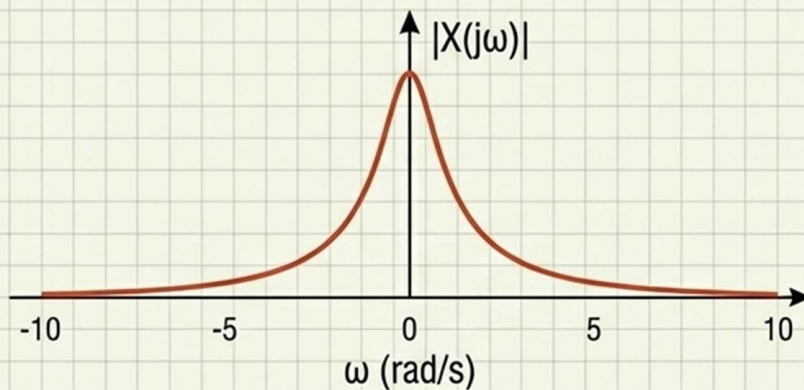
Operation	Mathematical Notation	Visual Effect	MATLAB Implementation
Shifting	$x[n-k]$	 Positive k delays, Negative k advances Time-shift	Altering the time array (e.g., $t - t_0$)
Scaling	$x[at]$	 $a > 1$ compresses, $0 < a < 1$ expands Time-scale	Multiplying the time array (e.g., $a * t$)
Reversing	$x[-n]$	 Flipped across Y-axis Time-reverse	Negating the time array (e.g., $-t$)

Core Operations Matrix

	1. Shifting	2. Scaling	3. Reversing
Mathematical Definition	$y[n] = x[n-k]$	$y(t) = x(at)$	$y[n] = x[-n]$
Visual Impact (Time Domain)			
Behavioral Logic	Positive k delays (moves right); Negative k advances (moves left).	$a > 1$ compresses time; $0 < a < 1$ expands time.	Flips the timeline entirely across $n=0$; the past becomes the future.

The Frequency Domain Crossing

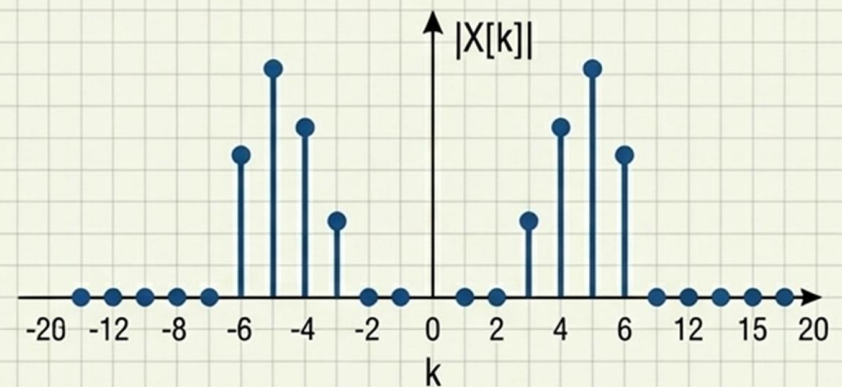
Continuous-Time Fourier Transform (CTFT)



$$x(t) = e^{(-\alpha * t)} * u(t)$$

$$X_{jw} = @(w) 1 ./(\alpha + 1j * w);$$

Discrete-Time Fourier Series (DTFS)

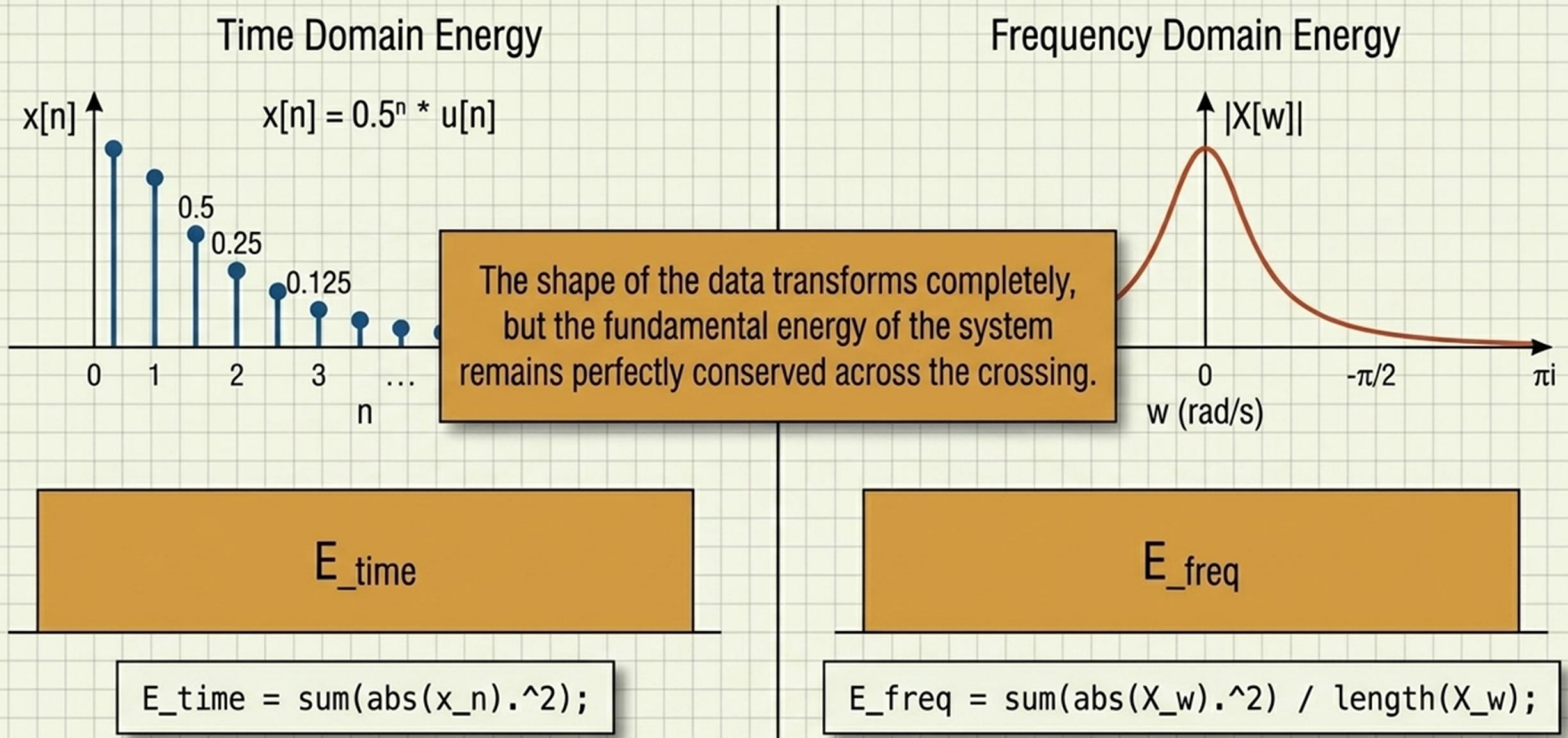


Periodic sequence cos + sin (N=20)

$$X_{dtfs} = \text{fft}(x_{dtfs}, N);$$

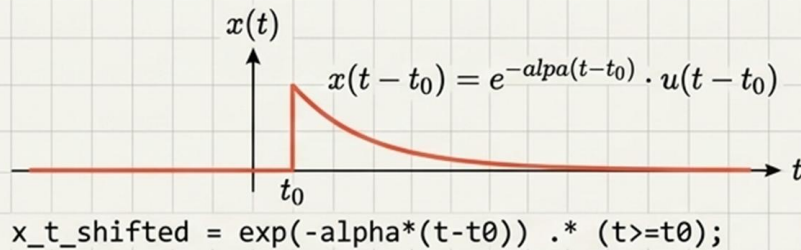
The bridge between domains: Time-domain signals map to frequency-domain spectra.

Parseval's Theorem: The Conservation of Energy



Duality in Action: Time Shift vs. Phase Shift

Cause (Time Domain)

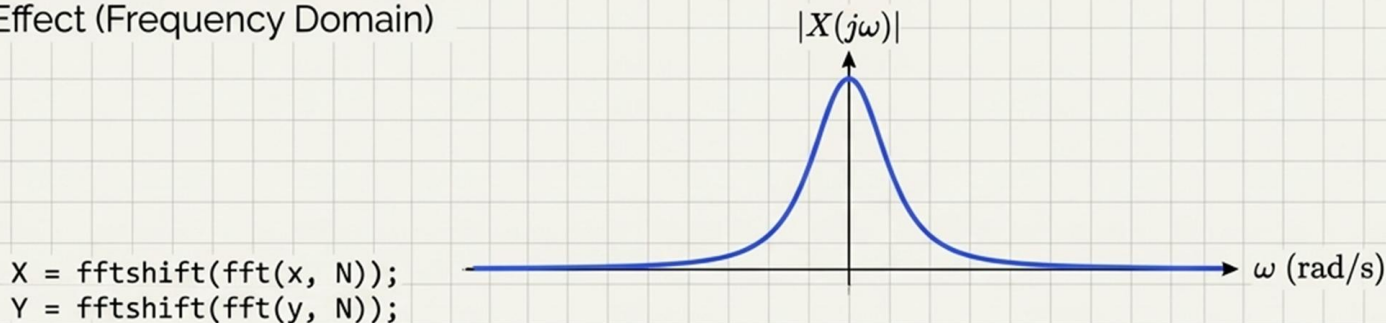


$$x(t - t_0) \longleftrightarrow X(j\omega)e^{-j\omega t_0}$$

A shift in the time domain alters the **phase** spectrum while leaving the magnitude spectrum unchanged.

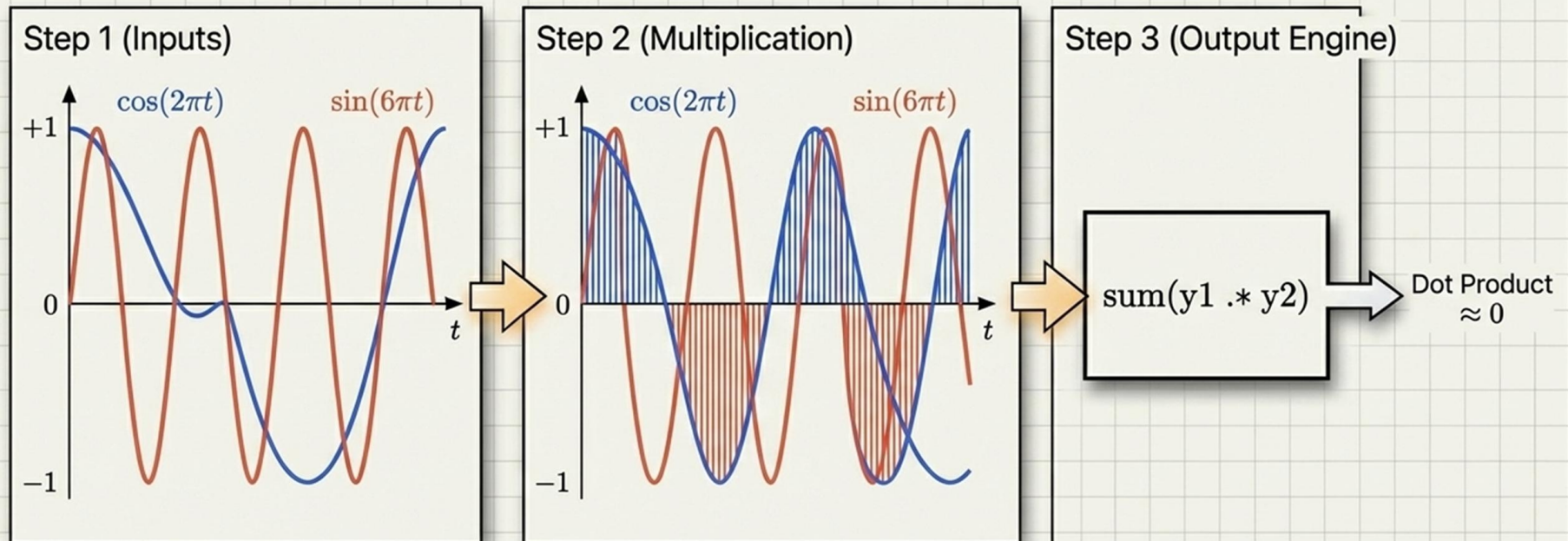


Effect (Frequency Domain)



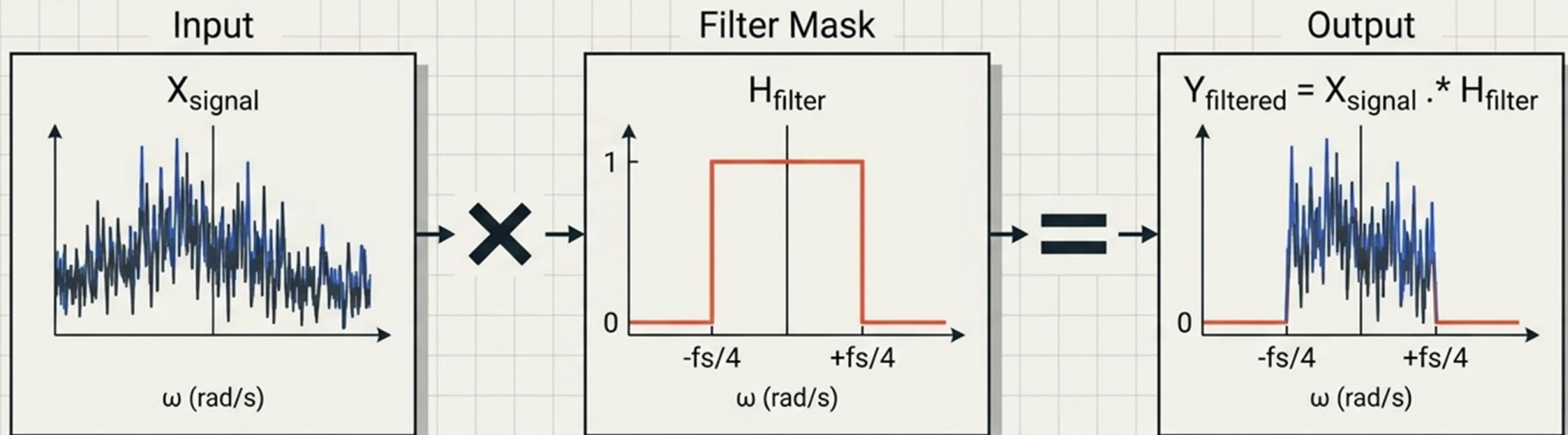
The Orthogonality Engine

Frequencies that are integer multiples of each other have zero net overlap.
They exist **entirely independent** of one another.



Filter Anatomy: Frequency Domain Operations

Filtering in the Frequency Domain is purely multiplicative. A low-pass filter acts as a physical gate, cutting off any frequencies outside the designated threshold.



User Inputs



Type of Operation:



Amount (+ or -)



The 4 Signals Challenge



1x Sinusoidal wave
(sine or cosine)



1x Sawtooth or
Square signal



2x Custom signals
of the student's
choice



Implementing using
App Designer

System Constraints

WARNING

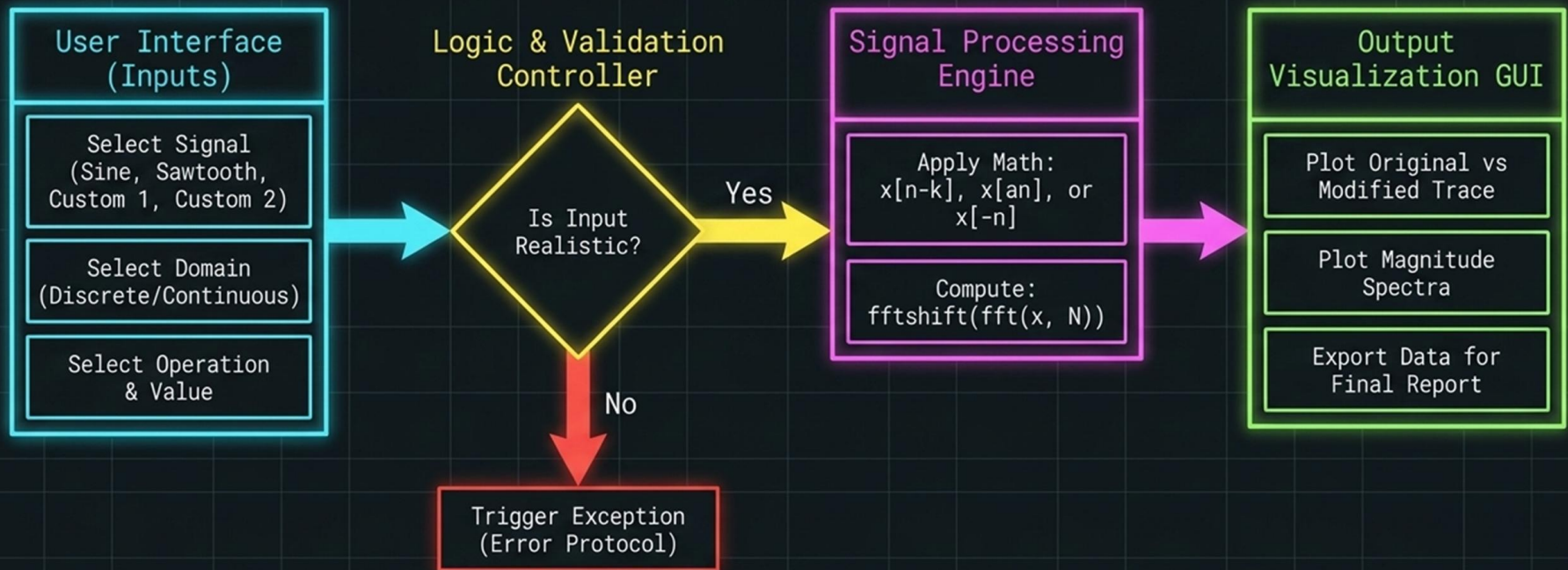
Error Handling:

Exception should be
thrown in case the user
inputs an unrealistic
value



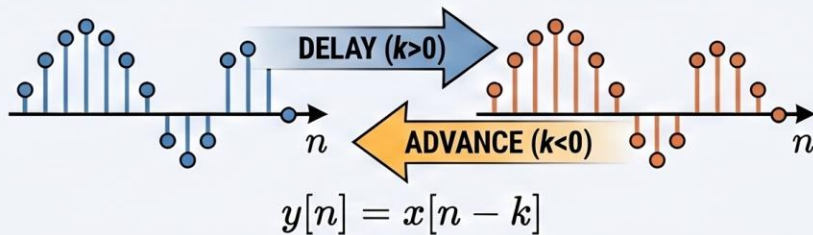
Reporting: Write a
comprehensive
report explaining
each step of the
MATLAB algorithm
implementation.

Synthesis: The App Designer Pipeline



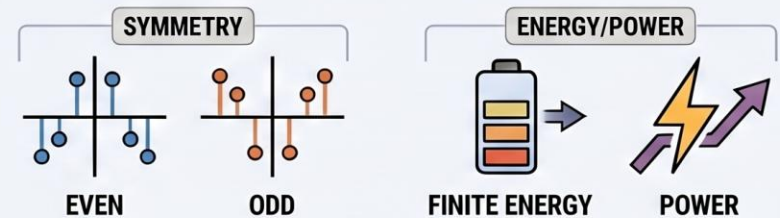
Discrete-Time Signal Manipulation & Analysis

1 Master Time-Shifting Fundamentals



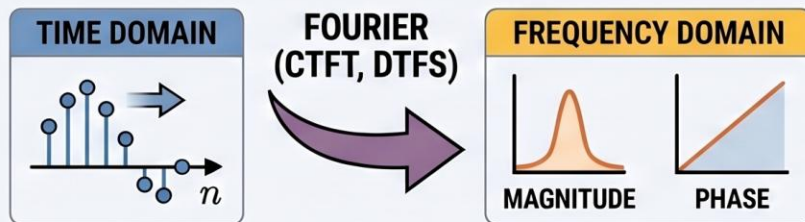
Use to **delay** (positive k) or **advance** (negative k) signals for applications like echo generation.

2 Analyze Symmetry, Energy, and Power



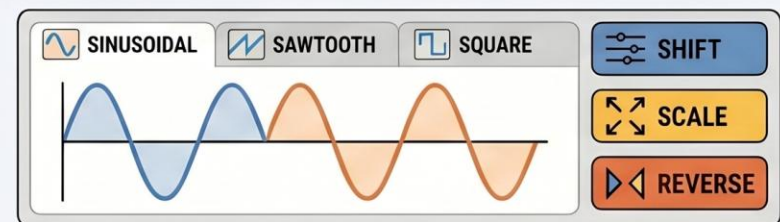
Define signals by periodicity, symmetry (even/odd), and whether they have **finite energy** or **power**.

3 Bridge Time and Frequency Domains



Use Fourier Representations to visualize how time-shifts affect phase and magnitude spectra.

4 Implement via MATLAB App Designer



Program dynamic tools to perform shifting, scaling, and reversing on sinusoidal, sawtooth, and square waves.



Jump Together, Fly Farther!

Next Steps: Complete your MATLAB GUI implementations
and finalize the step-by-step explanatory report.

Prof. Mehdi Pirahandeh
E-mail: mehdi@inha.ac.kr

